STANDATA interpretation 23-BCI-042/20-PCB-011R1

Building/Plumbing

Hydronic heating systems

Date Issued: July 2025

Purpose

This interpretation clarifies the design requirements of hydronic heating systems, so an acceptable level of performance may be maintained for all installations.

Discussion

The National Building Code – 2023 Alberta Edition (NBC(AE)) prescribes heating and air-conditioning systems must be designed, constructed and installed to conform with "good practice." While there are publications by industry and standards development organizations that are recognized by authorities having jurisdiction (AHJ) for use as "good practice," safety codes officers (SCO) have asked for guidance regarding the process for verifying hydronic heating system has been designed, constructed, and installed according to the "good practice" documents referenced in the NBC(AE)

For buildings within the scope of Division C, Sentence 2.4.1.1.(3) identifies full professional involvement is required on the project and, in the specific case of hydronic heating systems for these buildings, a registered engineering professional is usually retained to do the design and field review.

Buildings outside the scope of Division C, Sentence 2.4.1.1.(3) are not required to have professionals involved in the design and field review process, unless the SCO is of the opinion that some or all aspects of the building design are complex enough to warrant professional involvement. For these buildings, SCOs can use this interpretation as a guide to evaluate and review the design of hydronic heating systems submitted for a building permit.

Code References

Article 9.33.4.1.

9.33.4.1. Design of Heating and Air-conditioning Systems

1) Heating and air-conditioning systems, including ducting, and mechanical heating and refrigeration equipment, shall be designed, constructed and installed to conform with good practice such as that described in

a) the ASHRAE Handbooks and Standards,
b) the HRAI Digest,
c) the CHC Handbook on Hydronic Heating Systems,
d) the Hydronics Institute Manuals,
e) the SMACNA Manuals, and
f) CSA B214, "Installation code for hydronic heating systems."

(See also Subsection 9.32.3. for the design of systems that also provide ventilation.)

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Division C, Sentence 2.4.1.1. (3)

2.4.1. Professional Involvement
2.4.1.1. General
3) For *buildings* other than those described in Sentence (1), the drawings and specifications shall be authenticated by one or more *registered professionals* qualified to engage in the appropriate combination of those areas of the practice of architecture and engineering that are applicable to the project.

Application

This interpretation applies to any hydronic heating system intended to provide heat to the interior of the building. This includes systems designed as the primary heat source in the building, as well as any secondary heat sources (i.e. a house heated with a forced-air furnace having hydronic heating at the perimeter of a walk-out basement). Examples of hydronic heating systems would include, but not limited to:

1. Poured-floor radiant tubing
2. “Staple-up” radiant tubing
3. Convective-plate in-floor tubing
4. Radiators
5. Baseboard heaters
6. Fan-coil units

This interpretation does not apply to systems not intended to provide heat to the interior of the building, such as exterior snowmelt systems or pool heaters.

Interpretation

Proper verification of compliance with Sentence 9.33.4.1.(1) for hydronic heating systems in buildings not within the scope of Division C, Sentence 2.4.1.1.(3) can be obtained by the following.

DESIGN REQUIREMENTS

Hydronic heating systems must be:

1. Submitted as a pre-engineered package for review by the AHJ. Pre-engineered packages must consist of the basic generic system specifications and installation details prepared by a registered engineering professional together with additional system design data and floor plans specific to the project. The additional system design must be completed by

- A registered engineering professional licensed to practice in the province of Alberta; or
- An individual holding one of the following qualifications:
  - A Certified Hydronics Designer, certified by the Canadian Hydronics Council, a council within the Canadian Institute for Plumbing and Heating; or
  - A Residential Hydronics Design Technician, certified by the Heating, Refrigeration and Air Conditioning Institute of Canada.

2. Submitted as a custom design by

- a registered engineering professional licensed to practice in the province of Alberta; or
- an individual who holds one of the following qualifications:
  - A Certified Hydronics Designer, certified by the Canadian Hydronics Council; or
  - A Residential Hydronics Design Technician, certified by the Heating, Refrigeration and Air Conditioning Institute of Canada.

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SUBMISSION REQUIREMENTS

For all hydronic heating system designs, the plans and specifications submitted must include but not limited to the following information:

1. The schematic of the system and equipment specifications including, but not limited to, boilers, pumps, expansion tanks, zone controls, mixing valves and other system components such as supplementary baseboard and/or fan-coil units, water heater, etc. connecting to the system.
2. Mechanical room layout, venting and combustion air provisions for all fuel-fired appliances.
3. Piping specifications, spacing, sizes, maximum loop lengths, and pipe support details. Floor plans showing a general layout of the piping loops required for each room or space, and the location of the main headers, if applicable.
4. Locations, sizes and specifications for all heat terminal units, such as baseboard heaters, radiators, fan-coil units, etc., if applicable.
5. Cross sections through typical floor assemblies to show piping loop locations and the type of insulation to be provided.
6. System operating parameters including supply and return water temperatures, design flow rates and heat output coefficient of individual piping loops.
7. Room by room heat loss calculations.
8. Heat Exchangers.
9. Toxic Heat transfer fluids must be separated from the potable water using atmospherically vented double wall heat exchangers or equivalent protection.
10. Single wall heat exchangers may be installed when the heat transfer fluid is non-toxic and the installation complies with STANDATA 20-PCB-004 Single wall heat exchangers.

VENTILATION REQUIREMENTS

The NBC(AE) stipulates adequate ventilation be provided for all rooms and spaces in any building. Such provisions to be properly identified, in detail, on the system design drawings submitted for review.

Sections 6.2. and 9.32. of the NBC(AE) apply to the design requirements for ventilation systems in various applications.

This INTERPRETATION is applicable throughout the Province of Alberta.

Unless stated otherwise, all Code references in this STANDATA are to Division B of the National Building Code – 2023 Alberta Edition

Issued by the Provincial Building Administrator and Provincial Plumbing Administrator

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